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Studierichting: Informatica

Oefeningenreeks 1C nr. 31.5

Opgave: $(\sqrt{6} + i\sqrt{2})^{12}$ Bereken in $\mathbb{C}$ door gebruik van goniometrische vormen

$$\Rightarrow |z| = \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2} = \sqrt{6 + 2} = \sqrt{8} = 2\sqrt{2}$$

$$\tan(\theta) = \frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}$$

We weten dat $\tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$.

Dus $\theta = \frac{\pi}{6}$.

$$\Rightarrow z^{12} = (2\sqrt{2})^{12}(\cos(12 \cdot \frac{\pi}{6}) + i \sin(12 \cdot \frac{\pi}{6}))$$

$$= (2\sqrt{2})^{12}(\cos(2\pi) + i \sin(2\pi))$$

$$= 2^{\frac{3}{2} \cdot 12}(\cos(2\pi) + i \sin(2\pi))$$

$$= 2^{18}(\cos(2\pi) + i \sin(2\pi))$$

$$= 2^{18}$$

References